AI-POWERD CONGNITIVE SCREENING AND ANALYSIS PLATFORM : DEMENTIA SCREENING VIA SPEECH, FACIAL EXPRESSIONS, AND TASK BEHAVIOUR

Project ID: 25-26J-439

Project Proposal Report

N.Hashmath Al-Hazan

BSc (Hons) Degree in Information Technology Specializing in Software Engineering

Department of Computer Science & Software Engineering

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DECLARATION:

We declare that this is our own work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or institute of higher learning, and to the best of our knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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25/08/2025 Date

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List of Abbreviations

AI	Artificial Intelligence
ML	Machine Learning
IoT	Internet of Things
CMP	Contextual Matrix Profile
BPSD	Behavioral and Psychological Symptoms of Dementia
API	Application Programming Interface
MERN	MongoDB, Express.js, React.js, Node.js
WBS	Work Breakdown Structure
NGO	Non-Governmental Organization
HTTPS	Hypertext Transfer Protocol Secure

ABSTRACT

Dementia is a progressive neurological disorder that impairs memory, communication, and behavior, with early detection being essential for effective intervention and improved patient outcomes. This project component proposes a Multimodal dementia screening system that integrates speech analysis, facial expression recognition, and behavioral pattern monitoring to identify early signs of cognitive decline. The system leverages patient voice recordings, facial Action Units (AUs), blink rate, head motion, and task-interaction metrics such as response delays, repetitions, and hesitation markers to capture subtle cognitive deviations.

The methodology involves preprocessing audio and video data, extracting linguistic, acoustic, and facial-behavioral features, and applying both supervised (Random Forest, Logistic Regression) and unsupervised (Isolation Forest) machine learning models to generate a Cognitive Deviation Score. This score quantifies abnormal patterns relative to healthy controls, enabling risk classification into mild cognitive impairment or dementia stages. To ensure interpretability, explainable AI techniques (e.g., SHAP, LIME) are employed to highlight the most significant speech or facial features contributing to the classification.

Anticipated results include robust accuracy in distinguishing dementia stages, early detection of high-risk individuals through combined voice and facial cues, and improved accessibility to non-invasive screening outside clinical environments. The proposed solution is designed to be lightweight, privacy-preserving, and user-friendly, making it suitable for both home-based and clinical deployment.

In conclusion, this component addresses critical gaps in current dementia detection methods by fusing multimodal biomarkers into a single screening platform. The expected outcome is a scalable, real-time system that empowers individuals and caregivers with proactive insights, ultimately contributing to better dementia risk management and quality of life.

Keywords: Dementia Screening, Voice Recognition, Facial Expression Analysis, Behavioral Patterns, Multimodal AI, Explainable Machine Learning

1. INTRODUCTION

Dementia is a progressive neurological condition that affects millions of individuals worldwide, gradually impairing memory, communication, and behaviour. According to the World Health Organization, more than 55 million people currently live with dementia, with nearly 10 million new cases diagnosed annually. Alzheimer's disease is the most common subtype, accounting for 60–70% of cases, though other forms such as vascular dementia and frontotemporal dementia also contribute significantly to the global burden. The prevalence of dementia is expected to rise sharply with ageing populations, posing a critical challenge to healthcare systems and caregivers.

Early detection is vital for dementia management, as timely diagnosis enables effective intervention, lifestyle adjustments, and therapeutic strategies that can delay progression and improve quality of life. However, conventional diagnostic approaches, including neuroimaging (MRI, PET), clinical evaluations, and cognitive tests such as the Mini-Mental State Examination (MMSE), often detect dementia only at moderate to advanced stages. These methods are resource-intensive, require specialist supervision, and are not easily accessible in low-resource or home-based settings. As a result, there is a growing demand for affordable, non-invasive, and scalable screening tools that can identify dementia in its early stages.

Recent advances in artificial intelligence (AI) and machine learning (ML) have opened new avenues for digital health screening. Studies have shown that speech patterns, facial expressions, and behavioural responses can serve as sensitive biomarkers of cognitive decline. Dementia patients frequently display slower speech, increased pauses, reduced lexical diversity, abnormal blink rates, diminished emotional expressivity, and repetitive task behaviours. These subtle markers are difficult for caregivers to monitor consistently but can be captured and quantified using modern AI-driven techniques.

Despite promising progress, existing systems tend to focus on single modalities such as speech or facial cues in controlled environments, limiting their effectiveness in real-world applications. Few solutions integrate multimodal features—combining voice, facial expressions, and behavioural patterns—to provide a holistic and accurate assessment. Furthermore, most models act as "black boxes," offering little insight into which features drive predictions, which reduces clinical trust and adoption.

This project component addresses these gaps by proposing a Multimodal dementia screening system that leverages voice analysis, facial expression recognition, and behavioural pattern monitoring. By fusing these digital biomarkers, the system aims to deliver a Cognitive Deviation Score that quantifies early cognitive risks in a transparent, accessible, and non-invasive manner. Designed for deployment on mobile and web platforms, the proposed solution will empower caregivers, clinicians, and families to proactively monitor cognitive health and intervene earlier, ultimately improving dementia care and patient outcomes.

1.1 BACKGROUND AND LITERATURE

Dementia is a progressive neurological condition that impairs cognitive functions such as memory, attention, communication, and problem-solving. It affects more than 55 million people worldwide, with nearly 10 million new cases diagnosed annually, and the prevalence is expected to rise sharply as global populations age [1]. Alzheimer's disease is the most common subtype, accounting for 60–70% of cases, while other forms include vascular dementia, Lewy body dementia, and frontotemporal dementia. The World Health Organization recognizes dementia as one of the major causes of disability and dependency among older adults, with significant social, emotional, and financial impact on patients, families, and healthcare systems [2].

Early detection of dementia is crucial, as timely intervention can delay progression, improve symptom management, and enhance quality of life. However, conventional diagnostic approaches such as neuroimaging (MRI, PET scans) and standardized cognitive tests (e.g., Mini-Mental State Examination) are resource-intensive, invasive, and often detect impairment only after substantial decline has already occurred [3]. This has created an urgent need for non-invasive, affordable, and accessible screening tools capable of detecting subtle changes at early stages.

Recent research indicates that everyday behaviours—such as speech patterns, facial expressions, and simple cognitive task performance—contain measurable biomarkers of cognitive decline. Dementia patients often exhibit reduced lexical diversity, increased speech pauses, frequent use of fillers, abnormal blink rates, diminished emotional expressivity, and repetitive or delayed task responses [4], [5]. Studies using video analysis have demonstrated that facial Action Units (AUs) and head movement patterns can differentiate dementia patients from healthy controls. For example, Okunishi et al. showed that dementia and MCI patients displayed fewer positive emotions and more negative affect during natural conversation, achieving an AUC of 0.933 in classification tasks [6]. Similarly, Zheng et al. demonstrated that facial features such as AUs, blink rate, and facial mesh patterns can be used to identify dementia, with deep CNN models reaching 92.6% accuracy [7].

In parallel, automatic speech and transcript-based analysis has gained attention. The DementiaBank and ADReSS challenge datasets demonstrate that linguistic features such as type-token ratio, filler frequency, and repetitions can be strong predictors of dementia stages [8]. A study published in *Aging* confirmed a strong correlation ($r = 0.834$) between facial expression recognition outputs and behavioural and psychological symptoms of dementia, supporting the value of digital facial-behavioural biomarkers [9].

Despite these advances, most existing systems focus on single modalities (only speech or only facial features) and are typically validated in controlled laboratory settings. Few solutions integrate facial expression cues with behavioural task performance into a single framework, limiting their applicability in real-world home or community-based settings. Furthermore, many deep learning models lack transparency, making clinical interpretation and adoption more difficult [10].

This project component builds on these findings by proposing a multimodal, non-invasive dementia screening system that combines speech transcripts, facial expression analysis, and behavioural task metrics. By extracting features such as blink frequency, head movement, AU activations, speech hesitations, and repeated task behaviours, the system generates a Cognitive Deviation Score that quantifies deviations from healthy cognitive performance. Unlike traditional diagnostic approaches, this screening tool emphasizes accessibility, affordability, and ease of use, supporting both home-based and clinical applications.

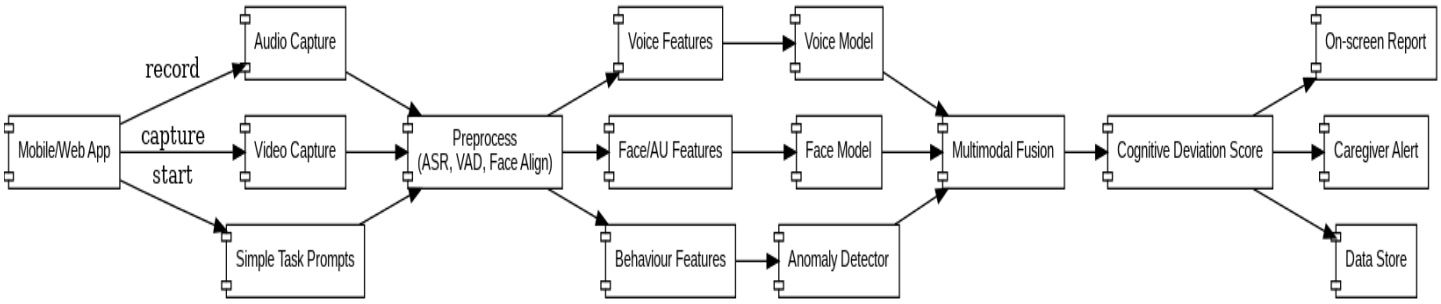


Figure 1-component diagram

1.2 RESEARCH GAP

When analyzing past research on dementia screening using speech, facial expression recognition, and multimodal behavioural features, six notable studies were reviewed. Each contributed toward advancing non-invasive dementia detection, yet several key limitations were identified, particularly in multimodal integration, explainability, affordability, and applicability in real-world settings.

Research A – Linguistic Features for Alzheimer's Detection [5]
This study used narrative speech transcripts from DementiaBank to predict Alzheimer's disease based on linguistic markers such as type-token ratio, filler frequency, and repetitions. While it achieved strong predictive power, it focused solely on speech, without integrating facial or behavioural features that could provide a more holistic assessment.

Research B – Automated Facial Expression Recognition for Dementia Screening [6]
A facial Action Unit (AU)-based recognition system was developed to distinguish dementia patients from healthy controls. Although the system achieved high classification accuracy (>90%), it relied on facial cues alone and did not include speech or cognitive task performance metrics, limiting its generalizability.

Research C – Multimodal Capture of Patient Behaviour [7]
This clinical feasibility study combined facial expressions, gaze, and task performance data for early dementia detection. While promising, it was tested in a controlled clinical environment rather than home-based or everyday settings, raising concerns about scalability and real-world usability.

Research D – Predicting Neuropsychiatric Symptoms Using FERS [8]
This study employed a Facial Expression Recognition System (FERS) to predict behavioural and psychological symptoms of dementia (BPSD). Although it demonstrated a strong correlation ($r = 0.834$) between facial affect and BPSD, it did not integrate behavioural task data or provide a unified scoring mechanism across modalities.

Research E – Dementia Detection Using HOG and Deep CNNs [9]
Facial mesh and handcrafted features (HOG) were compared with deep CNN models (Xception) for dementia detection using the PROMPT dataset. While CNNs achieved 92.6% accuracy, the study lacked explainability and offered limited insight into which features contributed most to predictions, hindering clinical trust.

Research F – Cookie Theft Picture Description Analysis [10]
This research applied linguistic and behavioural analysis to picture-description tasks to identify cognitive impairment. Although effective in identifying speech disfluencies and repetitions, it ignored facial dynamics (e.g., blink rate, head motion), which are equally relevant biomarkers.

Feature / Gap Dimension	Multimodal Input (Voice + Facial + Behaviour)	Task-based Cognitive Screening	Use of Consumer Devices (Webcam/Mic)	Unified Screening Framework	Explainable Results (SHAP/LIME)
Research A – Linguistic Features for Alzheimer's Detection [5]	✗	✓	✗	✗	✗
Research B – Automated Facial Expression Recognition [6]	✗	✗	✗	✗	✗
Research C – Multimodal Capture of Patient Behaviour [7]	✓	✗	✗	✗	✗
Research D – Predicting Neuropsychiatric Symptoms (FERS) [8]	✗	✗	✗	✗	✗
Research E – Dementia Detection (HOG vs CNN) [9]	✗	✗	✗	✗	✗
Research F – Cookie Theft Task Analysis [10]	✗	✓	✗	✗	✗
Proposed System (Your Component)	✓	✓	✓	✓	✓

Table 1-research gap

1.3 RESEARCH PROBLEM

The research problem of this project is to determine the feasibility of using a multimodal, non-invasive screening system that combines voice, facial expressions, and behavioural task performance for early dementia risk detection. Specifically, the problem is to explore how subtle markers in speech (pauses, fillers, repetitions), facial features (Action Units, blink rate, head movement), and task behaviour (response delays, hesitation, repeated actions) can be fused into a Cognitive Deviation Score to provide an accessible and interpretable risk assessment. The study also aims to evaluate the practicality of deploying such a system using consumer-grade devices (webcam and microphone) in home and clinical environments, while addressing challenges related to scalability, usability, and explainability.

The research aims to address the following questions:

- Can multimodal input (voice, facial, behavioural features) be reliably used to screen for early dementia risk in non-clinical settings?
- What are the potential benefits and limitations of using lightweight machine learning models (e.g., Logistic Regression, Random Forest, Isolation Forest) for multimodal risk assessment?
- How effective is the integration of task-based behavioural metrics (response time, repetitions, hesitations) in improving the accuracy of dementia screening compared to speech or facial features alone?
- What are the usability, scalability, and privacy considerations when implementing a multimodal dementia screening system using everyday devices (webcam/microphone)?
- How can explainable AI techniques (SHAP, LIME) enhance clinical trust by highlighting which voice or facial-behavioural features contribute to cognitive risk predictions?

By addressing these questions, the research will contribute to the development of a low-cost, accessible, and explainable dementia screening solution. The proposed system emphasizes early detection, practical deployment, and caregiver support, offering a pathway toward proactive dementia risk management that can complement existing clinical approaches.

2. OBJECTIVES

2.2 MAIN OBJECTIVES

To design and develop a multimodal, non-invasive dementia screening system that integrates voice, facial expressions, and behavioural task performance to generate a Cognitive Deviation Score for early dementia risk detection, while ensuring usability, scalability, and clinical interpretability.

2.3 SUB OBJECTIVES

1. Collect and process multimodal input using consumer-grade devices (microphone and webcam) to capture speech, facial expressions, and task responses.
2. Extract speech-based features such as pauses, filler frequency, lexical diversity, and repetitions from patient transcripts.
3. Extract facial and temporal features such as Action Unit (AU) activations, blink frequency, head motion, and emotional valence shifts.
4. Extract behavioural task metrics including response delays, repeated attempts, and hesitation markers during cognitive tasks.
5. Apply lightweight machine learning models (e.g., Random Forest, Logistic Regression, Isolation Forest) to classify dementia risk levels and detect anomalies.
6. Fuse multimodal features into a unified Cognitive Deviation Score (0–100) that quantifies deviation from healthy cognitive performance.
7. Incorporate explainable AI techniques (e.g., SHAP, LIME) to highlight key voice, facial, and behavioural features influencing risk classification.
8. Enable caregiver alerts and reporting, ensuring timely intervention when high-risk patterns are detected.
9. Ensure accessibility and affordability by designing the system to run on everyday devices and making it suitable for both home and clinical environments.
10. Develop a simple web/mobile interface that provides dementia risk screening results and caregiver notifications in a user-friendly format.

3. METHODOLOGY

The proposed methodology for this research component involves a structured sequence of steps to achieve the stated objectives. The system will use consumer-grade devices such as microphones and webcams to collect speech, facial expressions, and behavioural task responses, process these signals using lightweight machine learning models, and generate a Cognitive Deviation Score that reflects dementia risk.

The first step is data collection, where multimodal input will be captured through a web/mobile interface. This includes:

- Speech recordings from cognitive tasks (e.g., picture description, word fluency).
- Facial video streams captured during the same tasks to track Action Units, blink frequency, and head motion.
- Task interaction data such as response delays, repeated attempts, and hesitation markers.
  Publicly available dementia datasets (e.g., DementiaBank, PROMPT) will also be incorporated to support model training.

Next, data preprocessing will be performed to remove noise and prepare the features. For speech, this includes automatic transcription (ASR), filler detection, and segmentation of pauses. For facial data, preprocessing involves face detection, alignment, and landmark extraction to measure Action Units and temporal changes. Task data will be cleaned and normalized into behavioural metrics.

The third step is feature extraction, where:

- Speech features (e.g., type-token ratio, filler count, pause length).
- Facial features (e.g., AU intensities, blink frequency, head motion).
- Behavioural features (e.g., response delay, repetitions). will be derived and combined into feature sets suitable for machine learning.

The fourth step is model training and testing. Two categories of models will be applied:

- Supervised learning models (Logistic Regression, Random Forest) to classify dementia risk levels across Healthy, MCI, and Dementia groups.
- Unsupervised learning models (Isolation Forest) to detect anomalies in multimodal patterns that deviate from normal cognitive behaviour.

Following model evaluation, the system will implement a fusion and scoring module. This module will integrate the outputs of the models into a unified Cognitive Deviation Score (0–100). The score will provide a transparent, explainable indicator of dementia risk by applying explainable AI (SHAP, LIME) to highlight which features most influenced each prediction.

Finally, the system will incorporate a caregiver notification and reporting mechanism. When significant cognitive deviations are detected, caregivers will be notified via the web/mobile interface. The platform will present results in a simple, elderly-friendly format, providing clear insights into risk levels and contributing factors.

3.1 FEASIBILITY STUDY

3.1.1 TECHNIAL FEASIBILITY

The technical feasibility of this project is high due to the availability of consumer-grade devices (webcams and microphones), open-access dementia datasets, and mature machine learning frameworks. Speech data can be collected through standard microphone input and processed using automatic speech recognition (ASR) APIs such as Google Speech-to-Text or open-source alternatives like Whisper. Facial expressions and behavioural patterns can be captured using a webcam and analyzed with libraries such as OpenFace, MediaPipe, or OpenCV for face detection, Action Unit extraction, and blink/head-motion tracking.

Machine learning models, including Logistic Regression, Random Forest, and Isolation Forest, will be implemented using scikit-learn and XGBoost for lightweight, interpretable classification and anomaly detection. Explainable AI (XAI) techniques, including SHAP and LIME, will be applied to provide transparency on feature contributions. Data handling and preprocessing will be supported by Python libraries such as Pandas and NumPy.

The web and mobile interface can be developed using cross-platform frameworks such as React/React Native or MERN stack, enabling multimodal input (audio, video, and task responses) directly through a browser or mobile device. Caregiver alerting and reporting can be integrated using Firebase Cloud Messaging or Twilio APIs for real-time notifications. Deployment can be managed through containerized services (Docker/Kubernetes), ensuring scalability and cross-platform accessibility.

3.1.2 SECHEDULE FEASIBILITY

Schedule feasibility refers to the project's ability to be completed within the allotted timeframe. For this component, feasibility will depend on the timely availability of dementia datasets, access to consumer-grade devices (webcam and microphone), and setup of machine learning frameworks.

Possible delays may arise from technical issues in data preprocessing, model training, or multimodal feature fusion. To minimize risks, a structured project schedule with defined milestones will be prepared, covering tasks such as data collection, feature extraction, model development, and interface design.

With proper planning and monitoring, the project can be completed within the expected timeframe.

3.1.3 ECONOMIC FEASIBILITY

This project is economically feasible as it uses low-cost consumer devices (webcams, microphones) and relies on open-source tools and datasets such as Python, scikit-learn, and OpenCV, minimizing expenses. Deployment through affordable cloud services ensures scalability, making it a cost-effective non-invasive screening tool suitable for wider adoption.

3.2 SYSTEM ANALYSIS

3.2.1 SOFTWARE SOLUTION APPROACH

- Data Collection:
  Capture multimodal input using consumer-grade devices such as microphones and webcams. Speech recordings will be collected during structured cognitive tasks (e.g., picture description, word fluency), while webcams will capture facial expressions, blink rate, and head movements. Behavioural task data such as response delays and repetitions will also be logged. Open-access dementia datasets (e.g., DementiaBank, PROMPT) will supplement training and validation.
- Pre-processing:
  Clean and process audio and video data to remove noise, segment input, and normalize features. Speech will undergo ASR transcription, pause/filler detection, and lexical analysis, while facial data will be pre-processed using face detection, alignment, and landmark extraction to measure Action Units (AUs) and temporal changes. Task performance data will be standardized for consistency.
- Feature Extraction:
  Derive speech features (e.g., filler count, type-token ratio, pause length), facial features (e.g., AU activations, blink frequency, head motion), and behavioural features (e.g., hesitation markers, response times). These features will form the basis for classification and anomaly detection.
- Model Training:
  Train lightweight supervised models (Logistic Regression, Random Forest) to classify dementia risk levels, and use unsupervised models (Isolation Forest) to detect anomalies in multimodal behaviour.
- Model Evaluation:
  Evaluate models using accuracy, precision, recall, and F1-score, supported by cross-validation to reduce overfitting and ensure generalization.
- Cognitive Deviation Score Generation:
  Fuse outputs from all models into a single Cognitive Deviation Score (0–100), quantifying deviation from healthy cognitive behaviour. Explainable AI methods (SHAP, LIME) will be applied to highlight key features influencing each prediction.
- Caregiver Alert System:
  Implement a notification mechanism to alert caregivers when risk scores exceed a critical threshold, ensuring timely intervention.
- Web/Mobile Application Integration:
  Develop a cross-platform interface that allows users to complete screening tasks, view their results, and enables caregivers to remotely monitor reports. The interface will provide clear, elderly-friendly visualizations of risk levels.
- User Experience Design:
  Ensure accessibility for older adults by integrating large fonts, simple navigation, clear icons, and minimal text complexity.
- System Deployment:
  Deploy the backend and models using containerized services (Docker/Kubernetes) on a secure cloud platform, enabling real-time processing, scalability, and reliable data storage.

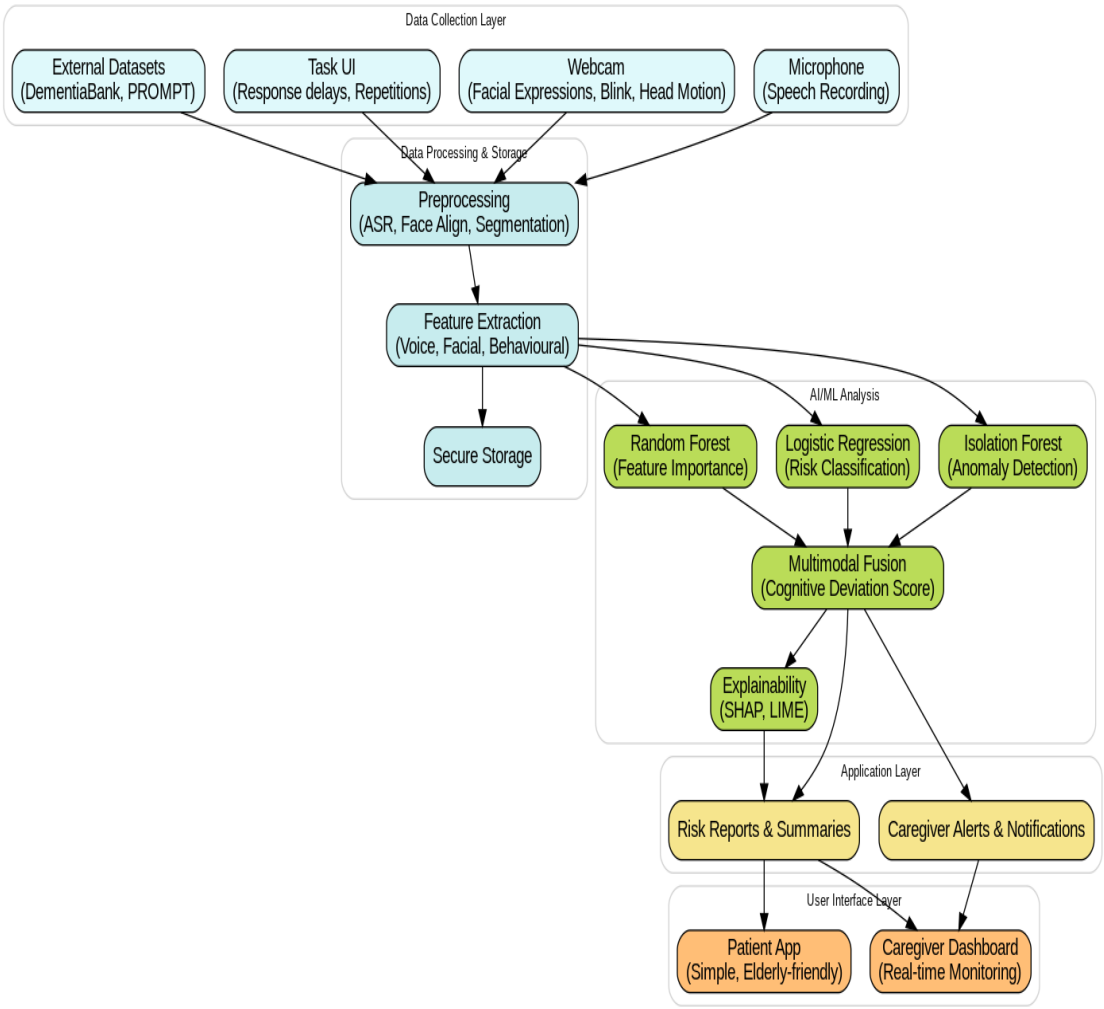


Figure 2-System Architecture diagram

3.2.2 TOOLS & TECHNOLOGY

Purpose	Technologies and Tools
Web/Mobile Application Development	React.js / Mern Stack, Express.js, Node.js
Machine Learning Frameworks & Algorithms	Supervised: Logistic Regression, Random Forest (classification)Unsupervised: Isolation Forest (anomaly detection) Explainability: SHAP, LIME Frameworks: scikit-learn, XGBoost
Speech Processing	Whisper (ASR), Google Speech-to-Text, NLTK, SpaCy
Facial Analysis	OpenCV, MediaPipe, OpenFace
Behavioural Task Logging	Custom Task Engine (built in React/Node)
Feature Extraction & Processing	Pandas, NumPy, SciPy
User Interface Development	React.js, Tailwind CSS, Chart.js (for reports/graphs)
Data Storage	MongoDB (session data), Cloud Storage (AWS/GCP for media)
Deployment	Docker, Kubernetes, Nginx
Security	HTTPS, JWT Authentication, Flask-Talisman
Version Control	Git, GitHub
Collaboration	Google Meet,WhatsApp
Project Management	Trello, Jira

Table 2-Tools

3.3 RESEARCH REQUIREMENTS

This project involves developing a non-invasive, multimodal dementia screening system that integrates speech, facial expressions, and behavioural task responses to generate a Cognitive Deviation Score for early dementia risk detection. The system will capture real-time audio and video data through consumer-grade devices such as microphones and webcams, supplemented by open-access dementia datasets.

The collected data will undergo preprocessing (speech transcription, facial alignment, temporal analysis) and feature extraction (pauses, fillers, Action Units, blink rate, response delays). Lightweight machine learning models such as Logistic Regression, Random Forest, and Isolation Forest will be applied for classification and anomaly detection. Model explainability will be incorporated using SHAP and LIME to provide transparent insights into feature importance.

The platform will include a web/mobile application where users can complete cognitive tasks, and caregivers can access screening results, reports, and alerts. A caregiver notification system will be integrated to ensure timely intervention when significant risk patterns are detected.

The project covers data collection, preprocessing, feature extraction, model training and evaluation, risk scoring, system integration, interface design, deployment, and usability testing. The system will be tested to ensure accuracy, interpretability, and accessibility.

The goal is to deliver a cost-effective, user-friendly, and scalable screening tool that supports early dementia detection and empowers caregivers with actionable insights.

3.3.1 FUNCTIONAL REQUIREMEMNTS

Functional requirements define the core features and capabilities that the system must provide to achieve its intended purpose. For this component, the functional requirements include:

- Data Collection: Capturing multimodal input using consumer-grade devices — microphone (speech recordings), webcam (facial expressions, blink rate, head motion), and task responses (reaction time, repetitions). Supplementary open datasets such as DementiaBank and PROMPT will also be used.
- Data Preprocessing: Processing raw audio and video, including speech transcription, voice activity detection, facial detection and alignment, and temporal analysis of blinks and head motion.
- Feature Extraction: Deriving measurable attributes from each modality — speech features (pauses, fillers, lexical diversity), facial features (Action Units, blink frequency), and behavioural features (response delay, hesitation markers).
- Model Training & Testing: Implementing lightweight supervised models (Logistic Regression, Random Forest) for dementia risk classification and unsupervised models (Isolation Forest) for anomaly detection.

- Cognitive Deviation Score: Integrating multimodal model outputs into a unified risk score (0–100) that quantifies deviations from healthy cognitive performance.
- Explainability Module: Applying SHAP and LIME to provide interpretable insights into which features most influenced dementia risk predictions.
- Caregiver Alert System: Generating notifications for caregivers when risk scores exceed defined thresholds or abnormal cognitive patterns are detected.
- Web/Mobile Application Interface: Providing an elderly-friendly interface for patients to perform screening tasks and caregivers to access reports, scores, and alerts.
- User Accessibility Features: Incorporating large fonts, simplified navigation, and intuitive icons to ensure ease of use for elderly individuals.

3.3.2 NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements describe the quality attributes and constraints that govern how the system should operate. For this component, they include:

- Performance: The system should process audio, video, and task responses in near real time to generate dementia risk scores and alerts without noticeable delays.
- Reliability: The system must consistently produce accurate and stable results when analyzing multimodal input, ensuring dependable dementia screening.
- Security: All sensitive audio/video data must be encrypted in transit and at rest, with strict access control to protect patient privacy.
- Scalability: The platform should support increasing numbers of users, concurrent video/audio uploads, and larger datasets without performance degradation.
- Usability: The interface should be simple, intuitive, and elderly-friendly, with large fonts, clear icons, and easy navigation for patients and caregivers.
- Availability: The system should maintain high uptime to ensure continuous screening and reliable caregiver notifications.
- Compatibility: The web/mobile application should run smoothly across major browsers and operating systems, supporting common consumer-grade devices (webcams, microphones).
- Maintainability: The architecture and codebase should allow easy debugging, modular updates, and integration of improved models or new datasets.
- Explainability: Model predictions must be interpretable using SHAP or LIME, allowing caregivers and clinicians to understand which features influenced risk assessment.

3.4 TEST

Testing will be a critical phase to ensure the reliability, accuracy, and usability of the proposed Multimodal Dementia Screening System. The initial plan is to conduct functionality testing within the development team and close associates, simulating real-world audio, video, and task-based data inputs. This stage will primarily involve unit testing and integration testing to verify that individual modules, such as data collection, preprocessing, feature extraction, and model inference, work correctly together.

In the final phase of the research, the system will be tested with a selected group of elderly participants and caregivers, first in a controlled environment and later in a real-world deployment setting. This stage will include acceptance testing, with Alpha testing to gather user feedback on usability and accessibility, followed by Beta testing to evaluate live performance, caregiver notifications, and explainability outputs.

Testing will include both manual and automated methods. Manual testing will verify user interface accessibility (large fonts, simple navigation), the accuracy of Cognitive Deviation Score presentation, and the effectiveness of caregiver alerts. Automated tests will use predefined dementia datasets and assert functions to validate the accuracy of machine learning models, consistency of multimodal fusion, and correctness of alert triggers. Special emphasis will be placed on ensuring that the system provides consistent results across different devices, browsers, and network conditions, while maintaining privacy and security of sensitive data.

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5. BUDGET AND BUDGET JUSTIFICATION

Description	Amount (LKR)
Webcam & Microphone (for multimodal data collection)	**10,000**
Cloud/Server Resources (for storage, preprocessing, model training & deployment)	**6,000**
Development Tools & Software (APIs, libraries, integration utilities)	**4,000**
Internet Cost (for data upload, testing, deployment access)	**2,000**
Miscellaneous (user testing support, printing, documentation)	**3,000**
Total	**25,000**

Table 3-Budget

6.WORK BREAKDOWN STRUCTURE

Phase	Task	Description
Phase 1: Planning	Requirement Gathering	Identify system requirements for smartwatch integration, ML model, and UI features
	Literature Review	Analyze existing systems and gaps
Phase 2: Data Handling	Data Collection	Gather smartwatch and public dataset data
	Data Preprocessing	Clean and prepare dataset for ML
Phase 3: Model Development	Feature Extraction	Extract relevant features (HRV, sleep metrics, activity patterns)
	Model Training & Testing	Train Logistic Regression and Isolation Forest models
Phase 4: System Integration	Backend Development	API integration for smartwatch data and AI models
	Frontend interface devlopmet	Build UI for elderly users and caregivers
Phase 5: Feature Implementation	Cognitive Tools Integration	Add music therapy, puzzles, and memory games
	Caregiver Alert Module	Implement real-time notifications
Phase 6: Testing & Deployment	System Testing	Functional, usability, and performance testing
	Final Deployment	Launch system on cloud

Table 4-WBS

7. GANTT CHART

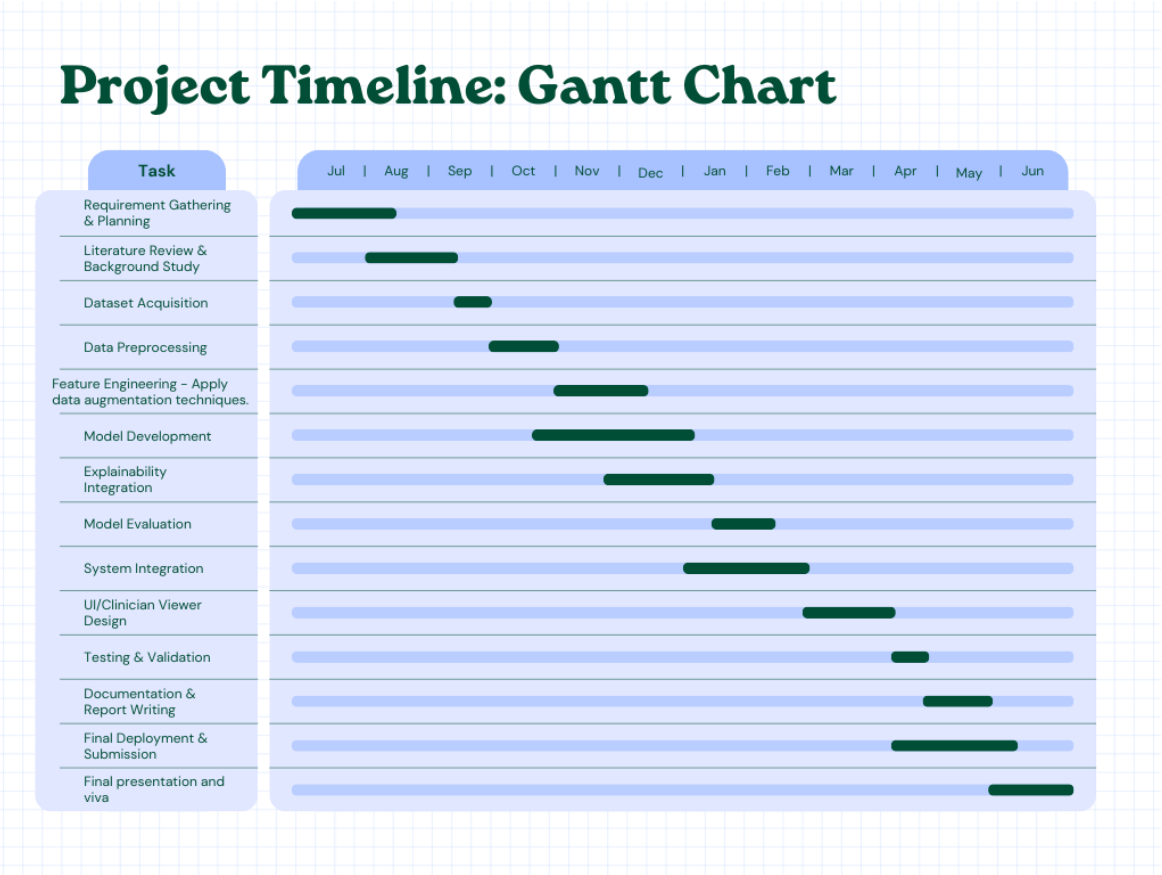


Table 5 - Gantt chart

8. COMMERCIALIZATION

The commercialization potential of this component lies in its ability to provide an affordable, non-invasive, and explainable dementia screening solution using widely available devices such as microphones and webcams. By enabling early detection of dementia risk through speech, facial expressions, and behavioural task analysis, the system addresses a pressing global healthcare challenge.

This solution can be marketed to elder care facilities, hospitals, memory clinics, NGOs, telehealth providers, and wellness organizations as a low-cost, AI-assisted screening and caregiver support platform.

It can be offered as:

- A subscription-based web/mobile application for families and caregivers to regularly screen elderly users at home.
- A white-labeled platform licensed to healthcare providers, insurance companies, and technology vendors for integration into their services.
- An integrated service embedded within existing telemedicine platforms, senior living centers, or national health monitoring programs.

The system's reliance on consumer-grade devices makes it highly scalable and cost-effective for both developed and developing countries. Strategic partnerships with government health agencies, telemedicine networks, and senior care organizations can accelerate adoption and broaden access.

Effective commercialization strategies will focus on:

1. Demonstrating healthcare savings through early detection, reducing costly hospitalizations and late-stage dementia treatments.
2. Emphasizing usability and accessibility, showing how elderly-friendly design and explainable outputs improve caregiver confidence and decision-making.
3. Leveraging digital marketing and B2B outreach to capture both consumer and institutional markets.

By addressing both direct-to-consumer and B2B healthcare segments, this component has the potential to become a sustainable, revenue-generating product that also delivers meaningful social impact by improving dementia care and early intervention.

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